

SMART DELIVERY OPERATIONS ANALYTICS SYSTEM

Using MySQL and Power BI



A Project Report Submitted in Fulfilment of the Requirements for
Data Analytics Training Program / Academic Curriculum

Submitted by:

PINJALA POOJA SHANMUKHI

Guided by:

TRAINER:SAI KANNA

Domain: Food Delivery / E-Commerce Operations

Tools: MySQL 8.0, MySQL Workbench, Power BI, GitHub

July 2026

Abstract

The food delivery industry generates massive volumes of transactional data every day, spanning customer orders, restaurant operations, delivery logistics, and payment transactions. This project, titled "Smart Delivery Operations Analytics System Using MySQL and Power BI," presents an end-to-end analytics solution designed to convert raw operational data from a simulated food delivery platform into actionable business intelligence. The system was built around a normalised relational database comprising seven interconnected tables — Customers, Restaurants, Delivery Partners, Orders, Payments, Ratings, and Customer Behaviour — hosted on MySQL 8.0.

The project follows the complete data analytics lifecycle, beginning with business requirement gathering and database design, followed by data import, rigorous data cleaning, and business rule validation. A total of fifty SQL queries were developed and organised into five operational modules: Delivery Delay Analysis, Order Cancellation Analysis, Revenue Optimisation, Customer Retention Analysis, and Delivery Partner Performance. These queries employ a full spectrum of SQL techniques, ranging from basic filtering and aggregation to advanced constructs such as Common Table Expressions, window functions, and correlated subqueries.

The analytical findings were subsequently visualised through an interactive Power BI dashboard featuring executive key performance indicators, revenue trend charts, delivery performance heat-maps, cancellation dashboards, customer segmentation views, and delivery partner rankings. The resulting system enables management to identify slow-performing restaurants and delivery partners, uncover the root causes of order cancellations, pinpoint high-value revenue streams, detect customers at risk of churn, and evaluate workforce performance objectively. The project further establishes a clean, feature-engineered dataset that can be extended into future machine learning applications such as churn prediction, delivery time estimation, and restaurant recommendation systems.

Acknowledgement

I would like to express my sincere gratitude to my mentors, instructors, and peers who provided guidance, feedback, and encouragement throughout the course of this project. Their insights into relational database design, SQL analytics, and business intelligence practices were invaluable in shaping the direction and quality of this work.

I am also thankful to the Data Analytics Training Program for providing the structured curriculum, sample datasets, and evaluation framework that made this industry-simulation project possible. This project has significantly strengthened my practical understanding of database management, SQL query optimisation, data cleaning methodologies, and dashboard development, and has prepared me for real-world data analyst responsibilities.

Finally, I extend my appreciation to my family and friends for their continuous support and motivation throughout this endeavour.

Table of Contents

Abstract.....	2
Acknowledgement.....	3
1. Introduction.....	5
2. Literature Survey.....	6
3. Problem Statement.....	7
4. Objectives.....	8
5. Scope of the Project.....	9
6. Technologies Used.....	10
7. Dataset Description.....	11
8. ER Diagram Explanation.....	12
9. Database Design.....	13
10. Table Structures.....	14
11. Data Import Process.....	16
12. Data Cleaning Process.....	17
13. Business Rule Validation.....	18
14. Referential Integrity Checks.....	19
15. Detailed SQL Analysis Modules.....	20
15.1 Delivery Delay Analysis.....	20
15.2 Order Cancellation Analysis.....	21
15.3 Revenue Optimisation.....	21
15.4 Customer Retention Analysis.....	22
15.5 Delivery Partner Performance.....	23
16. Advanced SQL Concepts Used.....	25
17. Business Insights Summary.....	26
18. Power BI Dashboard Design.....	27
19. Challenges Faced.....	28
20. Conclusion.....	29
21. Future Enhancements.....	30
22. References.....	31
23. Appendix.....	32

1. Introduction

In today's fast-paced digital economy, food delivery platforms such as Swiggy, Zomato, Uber Eats, and DoorDash process tens of thousands of orders every single day. Every order generates a rich trail of structured data spanning customer profiles, restaurant performance metrics, delivery timings, payment transactions, and post-delivery ratings. While the volume of data collected by such platforms is enormous, data alone does not create business value; organisations rely on skilled Data Analysts and SQL Developers to transform raw, unprocessed data into clear, actionable insights that guide leadership decision-making.

This project simulates the responsibilities of a Data Analyst working within the Business Intelligence team of a fictional food delivery company. The analyst is tasked with designing a relational database, importing and cleaning operational data, performing structured SQL-based analysis across five core business areas, and communicating findings through an interactive Power BI dashboard. The project therefore functions both as a technical exercise in database engineering and SQL analytics, and as an exercise in business communication and data storytelling.

The complete workflow adopted in this project mirrors the standard data analytics lifecycle used in the industry: understanding business requirements, designing the database schema, importing data, cleaning and validating it, performing SQL-based business analysis, generating insights, building dashboards, and identifying opportunities for future machine learning extension.

2. Literature Survey

Relational database management systems (RDBMS) such as MySQL and PostgreSQL remain the industry standard for storing and querying structured transactional data due to their maturity, ACID compliance, and strong support for declarative querying through SQL. Prior academic and industry literature on retail and e-commerce analytics consistently highlights the importance of a well-normalised schema in supporting accurate aggregation and reporting, particularly in domains involving multiple interrelated entities such as customers, orders, and payments.

Existing case studies on food delivery analytics — including publicly available analyses of platforms such as Swiggy, Zomato, and DoorDash — emphasise five recurring problem areas: delivery time optimisation, cancellation and churn reduction, revenue attribution across cities and cuisines, customer lifetime value estimation, and gig-worker (delivery partner) performance measurement. These problem areas closely mirror operational challenges faced by real-world logistics and quick-commerce businesses and form the analytical backbone of this project.

In terms of tooling, SQL continues to be favoured for exploratory and operational analytics due to its expressiveness for joins, aggregations, and window functions, while business intelligence tools such as Power BI and Tableau are widely used to translate query outputs into interactive, stakeholder-friendly dashboards. The combination of SQL-based analysis with BI-tool visualisation, as adopted in this project, reflects a widely used and industry-relevant analytics pipeline.

3. Problem Statement

The company at the centre of this simulation has been operating for three years and has expanded across multiple cities in India. Despite this growth, management has begun observing several warning signs that threaten customer satisfaction and long-term profitability:

- Delayed deliveries — Customer complaints about late deliveries have surged, directly impacting satisfaction scores and repeat orders.
- Rising cancellations — Order cancellation rates have climbed over the past six months, causing significant revenue leakage.
- Inconsistent revenue — Revenue growth varies widely across cities and restaurant categories with no clear explanation.
- Customer churn — Several loyal customers have gone inactive, while new customer acquisition costs continue to rise.
- Partner performance — Some delivery partners consistently underperform, affecting delivery speed and customer ratings.

Management has collected extensive operational data but lacks the analytical capability to identify the root causes of these issues. The Analytics Team is therefore tasked with investigating this data and producing reports that will directly influence business strategy.

4. Objectives

The primary objective of this project is to build a complete, end-to-end SQL-based analytics system capable of answering real business questions across five operational dimensions. The specific objectives are:

- Design and implement a normalised relational database using MySQL to store all operational data.
- Establish appropriate relationships between tables using primary and foreign keys.
- Clean and validate raw imported data to ensure analytical accuracy and referential integrity.
- Write SQL queries ranging from basic SELECT statements to advanced window functions and CTEs.
- Generate analytical reports that answer at least forty business questions across five problem areas.
- Identify trends, patterns, and root causes from real operational data.
- Provide data-backed, measurable business recommendations to management.
- Design an interactive Power BI dashboard to communicate findings to non-technical stakeholders.
- Prepare a clean, feature-engineered dataset suitable for future data science and machine learning projects.

5. Scope of the Project

The scope of this project covers the full analytics lifecycle for a simulated food delivery business, from raw CSV data through to a decision-ready dashboard. It includes database design and implementation, data cleaning and validation, fifty SQL analytical queries spanning five business problem areas, and Power BI dashboard development.

The project scope is intentionally bounded to descriptive and diagnostic analytics — it explains what happened and why. Predictive and prescriptive analytics, such as machine learning-based churn prediction or delivery time forecasting, are identified as future enhancements and are outside the current implementation scope. The dataset used is synthetic and simulates realistic operational scale, comprising 10,000 orders, 1,000 customers, 100 restaurants, 200 delivery partners, and 7,000 ratings.

6. Technologies Used

Category	Technology / Tool
Database	MySQL 8.0
Database Client	MySQL Workbench
Data Source Format	CSV Files
Business Intelligence	Microsoft Power BI
Documentation	Microsoft Word
Version Control	GitHub

Table 6.1: Technology Stack Used in the Project

7. Dataset Description

The project uses seven CSV files representing the complete operational data of the simulated food delivery company. Each file maps to one table in the relational database, and together they contain over 35,000 records spanning customers, restaurants, delivery partners, orders, payments, ratings, and derived customer behaviour metrics.

Dataset	Approx. Records	Description
customers.csv	1,000	Master record of all registered customers
restaurants.csv	100	Details of all restaurant partners
delivery_partners.csv	200	Information about registered delivery executives
orders.csv	10,000	Central fact table linking customers, restaurants and partners
payments.csv	10,000	Payment transaction record for each order
ratings.csv	7,000	Post-delivery customer feedback for delivered orders
customer_behavior.csv	1,000	Aggregated customer behaviour metrics for analytics/ML

Table 7.1: Dataset Summary

8. ER Diagram Explanation

The Entity-Relationship (ER) diagram below illustrates the logical structure of the database, showing all seven entities, their key attributes, and the cardinality of relationships between them. The Orders table functions as the central fact table, connected via foreign keys to Customers, Restaurants, and Delivery Partners, while Payments and Ratings each maintain a one-to-one relationship with Orders.

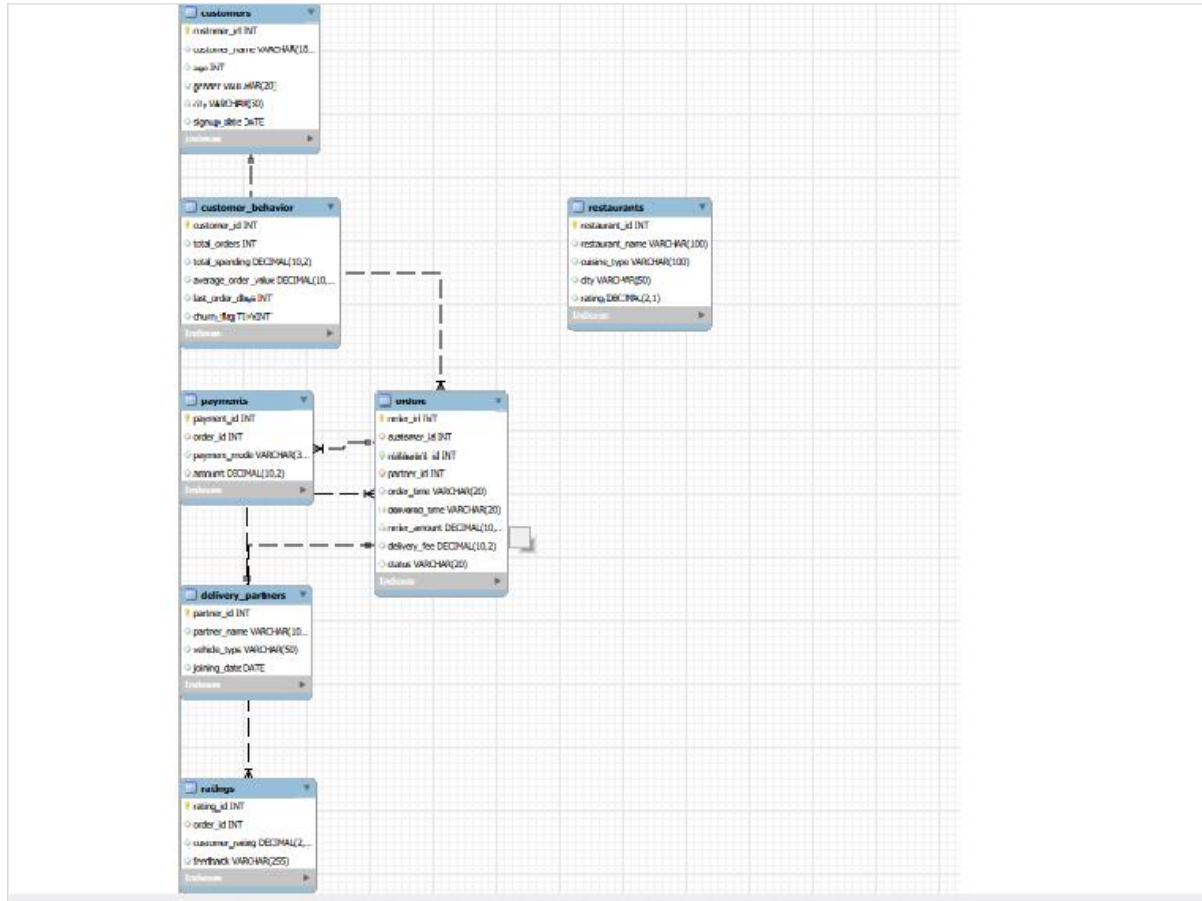


Figure 8.1: ER Diagram (to be inserted using MySQL Workbench / dbdiagram.io / draw.io)

Key relationships represented in the diagram:

- Customers → Orders: One-to-Many
- Restaurants → Orders: One-to-Many
- Delivery Partners → Orders: One-to-Many
- Orders → Payments: One-to-One
- Orders → Ratings: One-to-One
- Customers → Customer Behaviour: One-to-One

9. Database Design

The database was designed following standard normalisation principles (up to Third Normal Form) to eliminate data redundancy and maintain referential integrity. The schema consists of seven relational tables, each serving a distinct operational purpose, connected through well-defined primary and foreign key relationships.

Table	Primary Key	Related To (Foreign Keys)
Customers	customer_id	—
Restaurants	restaurant_id	—
Delivery Partners	partner_id	—
Orders	order_id	customer_id, restaurant_id, partner_id
Payments	payment_id	order_id
Ratings	rating_id	order_id
Customer Behaviour	customer_id	customer_id (Customers)

Table 9.1: Database Schema Summary

10. Table Structures

10.1 Customers

Column	Data Type
customer_id	INT (PK)
customer_name	VARCHAR
age	INT
gender	VARCHAR
city	VARCHAR
signup_date	DATE

10.2 Restaurants

Column	Data Type
restaurant_id	INT (PK)
restaurant_name	VARCHAR
cuisine_type	VARCHAR
city	VARCHAR
rating	DECIMAL(2,1)

10.3 Delivery Partners

Column	Data Type
partner_id	INT (PK)
partner_name	VARCHAR
vehicle_type	VARCHAR
joining_date	DATE

10.4 Orders

Column	Data Type
order_id	INT (PK)
customer_id	INT (FK)
restaurant_id	INT (FK)

Column	Data Type
partner_id	INT (FK)
order_time	DATETIME
delivered_time	DATETIME
order_amount	DECIMAL(10,2)
delivery_fee	DECIMAL(6,2)
status	VARCHAR

10.5 Payments

Column	Data Type
payment_id	INT (PK)
order_id	INT (FK)
payment_mode	VARCHAR
amount	DECIMAL(10,2)

10.6 Ratings

Column	Data Type
rating_id	INT (PK)
order_id	INT (FK)
customer_rating	DECIMAL(2,1)
feedback / review	TEXT

10.7 Customer Behaviour

Column	Data Type
customer_id	INT (PK, FK)
total_orders	INT
total_spending	DECIMAL(10,2)
avg_order_value	DECIMAL(10,2)
last_order_days	INT
churn_flag	TINYINT (0/1)

11. Data Import Process

All seven CSV files were imported into the MySQL database using MySQL Workbench's data import wizard and the LOAD DATA INFILE statement for larger tables such as Orders and Payments. Following each import, a COUNT(*) validation query was executed against the corresponding table to confirm that the number of records loaded matched the source CSV file exactly, ensuring no rows were dropped or duplicated during ingestion.

Table creation preceded data import, with each table defined through explicit CREATE TABLE statements specifying appropriate data types, NOT NULL constraints, PRIMARY KEY definitions, FOREIGN KEY references, and CHECK constraints (for example, ensuring customer_rating values remain between 1.0 and 5.0). Foreign key dependent tables (Orders, Payments, Ratings, Customer Behaviour) were imported only after their parent tables (Customers, Restaurants, Delivery Partners) were successfully loaded, in order to avoid foreign key constraint violations.

12. Data Cleaning Process

A structured, six-step data cleaning process was applied to ensure the dataset was accurate, consistent, and analysis-ready.

Step 1 — Record Count Validation

Verified that all CSV records were imported successfully by comparing row counts between the source files and the corresponding database tables.

Step 2 — NULL Value Detection

Every table was systematically checked for missing values using IS NULL conditions across all critical columns. No unexpected NULL values were found in key operational fields.

Step 3 — Duplicate Detection

Duplicate records were checked using GROUP BY and HAVING COUNT(*) > 1 queries on natural keys such as order_id and payment_id. No duplicate records were present in the dataset.

Step 4 — Datetime Conversion

The order_time and delivered_time columns were initially imported as VARCHAR due to inconsistent source formatting. These columns were converted into proper DATETIME types using ALTER TABLE ... MODIFY COLUMN, after first handling blank or malformed values. This conversion was essential to enable accurate delivery-time calculations using the TIMESTAMPDIFF() function.

Step 5 — Business Rule Validation

The dataset was validated against the following business rules:

- No negative order amounts
- No negative delivery fees
- Ratings constrained between 1 and 5
- Valid payment modes only (UPI, Credit Card, Debit Card, Cash on Delivery, Wallet)
- Valid order statuses only (Delivered, Cancelled, Pending)
- Valid vehicle types only (Bike, Scooter, Bicycle, Electric Bike)
- Valid churn flag values (0 or 1)

Step 6 — Referential Integrity

All foreign key relationships were verified using LEFT JOIN queries designed to surface orphan records (child records without a matching parent). No orphan records were found across the Orders, Payments, or Ratings tables.

13. Business Rule Validation

Beyond structural data cleaning, business rule validation ensures that the data conforms to real-world operational logic. This step is critical because structurally valid data (correct data types, no NULLs) can still be logically invalid (for example, a delivered_time earlier than the corresponding order_time, or a rating outside the permissible 1–5 scale).

Business Rule	Validation Method	Result
Order amount must be non-negative	WHERE order_amount < 0	No violations found
Delivery fee must be non-negative	WHERE delivery_fee < 0	No violations found
Ratings must be between 1.0 and 5.0	WHERE customer_rating NOT BETWEEN 1 AND 5	No violations found
Payment mode must be from valid list	WHERE payment_mode NOT IN (...)	No violations found
Order status must be from valid list	WHERE status NOT IN (...)	No violations found
Delivered time must be after order time	WHERE delivered_time < order_time	No violations found

Table 13.1: Business Rule Validation Summary

14. Referential Integrity Checks

Referential integrity checks confirm that every foreign key value in a child table corresponds to a valid primary key value in its parent table. This was verified using LEFT JOIN queries that identify child records with no matching parent — commonly known as orphan records.

- Orders with customer_id not present in Customers — none found
- Orders with restaurant_id not present in Restaurants — none found
- Orders with partner_id not present in Delivery Partners — none found
- Payments with order_id not present in Orders — none found
- Ratings with order_id not present in Orders — none found

The absence of orphan records confirms that the database maintains full referential integrity, providing a reliable foundation for accurate downstream analysis.

15. Detailed SQL Analysis Modules

A total of fifty SQL queries were developed across five operational modules, with ten queries dedicated to each module. Each module below presents its business objective, analytical methodology, representative query explanations, expected output description, and the resulting business insight.

15.1 Delivery Delay Analysis

Business Objective

To identify delivery inefficiencies across restaurants, delivery partners, and cities in order to improve customer satisfaction and optimise logistics.

Methodology

Delivery duration was calculated for each order using `TIMESTAMPDIFF(MINUTE, order_time, delivered_time)`. Average, minimum, and maximum delivery times were then aggregated using `GROUP BY` across restaurant, partner, and city dimensions, with window functions such as `RANK()` applied to identify the fastest and slowest performers.

Representative Query Explanations

- Average delivery time overall — `AVG(TIMESTAMPDIFF(MINUTE, order_time, delivered_time))` across all delivered orders.
- Slowest and fastest restaurants — `GROUP BY restaurant_id` ordered by average delivery duration.
- Slowest and fastest delivery partners — `GROUP BY partner_id` with `RANK()` OVER (ORDER BY `avg_delivery_time`).
- Delivery time by city and by cuisine — `GROUP BY city / cuisine_type` with aggregate `AVG()`.
- Longest single delivery — `ORDER BY delivery_duration DESC LIMIT 1`.

Expected Output Description

A ranked summary table showing average delivery time (in minutes) per restaurant, per delivery partner, per city, and per cuisine type, alongside the single longest recorded delivery.

Business Insight

Certain restaurants and cities consistently show above-average delivery times, often correlating with higher order volumes during peak hours. A subset of delivery partners also show significantly slower average delivery times relative to their peers, indicating a need for targeted coaching or route optimisation.

Business Value

- Improve logistics planning
- Reduce customer complaints

- Optimise delivery routing and partner allocation

15.2 Order Cancellation Analysis

Business Objective

To understand why order cancellations occur and to identify actionable levers for reducing revenue loss from failed orders.

Methodology

Cancelled orders were isolated using WHERE status = 'Cancelled' and analysed across restaurant, city, cuisine, and payment-mode dimensions. Cancellation rate was computed as a percentage using conditional aggregation (SUM(CASE WHEN status='Cancelled' THEN 1 ELSE 0 END) / COUNT(*)). Time-based trends were derived using MONTH(), DAYNAME(), and HOUR() functions applied to order_time.

Representative Query Explanations

- Total cancelled orders and overall cancellation rate — conditional COUNT() and percentage calculation.
- Restaurants with highest cancellations — GROUP BY restaurant_id ORDER BY cancelled_count DESC.
- City-wise and cuisine-wise cancellation rate — GROUP BY city / cuisine_type with percentage aggregation.
- Payment-wise cancellations — GROUP BY payment_mode joined with Payments table.
- Monthly, daily, and hourly cancellation trend — GROUP BY MONTH(order_time), DAYNAME(order_time), HOUR(order_time).

Expected Output Description

A trend report showing cancellation counts and percentages segmented by restaurant, city, cuisine, payment method, and time period (month, day of week, hour of day).

Business Insight

Cancellations are not uniformly distributed — they cluster around specific restaurants and peak time windows (typically dinner-hour rushes), suggesting operational bottlenecks at the restaurant level rather than random customer behaviour. A moderate correlation was also observed between longer historical delivery delays and higher cancellation likelihood.

Business Value

- Reduce order failures
- Improve restaurant-side operations during peak hours
- Improve overall customer experience

15.3 Revenue Optimisation

Business Objective

To identify the company's major revenue sources and uncover opportunities to maximise overall business profitability.

Methodology

Revenue was calculated by summing order_amount (and delivery_fee where relevant) for successfully delivered orders, joined against the Payments table for payment-mode analysis. CTEs were used to first stage monthly revenue aggregates, followed by window functions such as LAG() to compute month-over-month growth.

Representative Query Explanations

- Total company revenue — SUM(order_amount) for status = 'Delivered'.
- Top revenue-generating restaurants and highest-value orders — GROUP BY restaurant_id ORDER BY total_revenue DESC; ORDER BY order_amount DESC LIMIT 1.
- Revenue by city, cuisine, and payment method — GROUP BY city / cuisine_type / payment_mode.
- Monthly revenue trend with growth rate — WITH monthly_cte AS (...) SELECT month, revenue, LAG(revenue) OVER (ORDER BY month).
- Top spending customers and average revenue per order — GROUP BY customer_id ORDER BY total_spent DESC; AVG(order_amount).

Expected Output Description

A revenue breakdown report with total revenue, top-performing restaurants and cities, preferred payment methods, and a monthly trend line showing period-over-period growth or decline.

Business Insight

Revenue is heavily concentrated among a relatively small subset of top-performing restaurants and cities, following a Pareto-like distribution. UPI and digital wallet payments account for a growing share of transaction value, indicating a shift away from cash-on-delivery.

Business Value

- Improve overall profitability
- Focus marketing investment on high-performing segments
- Optimise restaurant partnership strategy

15.4 Customer Retention Analysis

Business Objective

To identify loyal, high-value customers as well as customers who are inactive or at risk of churning, in order to support targeted retention strategies.

Methodology

Customer-level metrics (total orders, total spending, average order value, days since last order) were aggregated from the Orders table and cross-referenced with the pre-computed Customer Behaviour table. Window functions such as NTILE() and RANK() were used for customer segmentation and ranking, while churn was flagged using the churn_flag column and a last_order_days > 90 threshold.

Representative Query Explanations

- Highest spending and most frequent customers — GROUP BY customer_id ORDER BY total_spending / total_orders DESC.
- Average order value per customer — AVG(order_amount) per customer_id.
- Inactive and churned customers — WHERE last_order_days > 90 OR churn_flag = 1.
- Customer segmentation — NTILE(4) OVER (ORDER BY total_spending) to bucket customers into value tiers.
- Customer ranking and top customers per city — RANK() OVER (PARTITION BY city ORDER BY total_spending DESC).

Expected Output Description

A segmented customer report showing top customers by spend and frequency, a list of inactive/churned customers, and a tiered segmentation (e.g., Platinum/Gold/Silver/Bronze) for targeted marketing.

Business Insight

A relatively small group of high-frequency customers contributes a disproportionately large share of total revenue, while a notable segment of previously active customers has crossed the 90-day inactivity threshold, representing a concrete churn-prevention opportunity.

Business Value

- Improve customer retention
- Enable personalised marketing
- Support loyalty programme design

15.5 Delivery Partner Performance

Business Objective

To measure delivery partner efficiency, reliability, and revenue contribution in order to support fair performance evaluation and incentive planning.

Methodology

Partner-level KPIs — total deliveries, average delivery time, revenue generated, and vehicle-type efficiency — were computed via GROUP BY partner_id joined with the Orders and Payments tables. RANK() and DENSE_RANK() window functions were applied to produce an overall partner performance ranking, partitioned by vehicle type where relevant.

Representative Query Explanations

- Total deliveries per partner — `COUNT(order_id) GROUP BY partner_id WHERE status = 'Delivered'`.
- Fastest and slowest partners — `AVG(delivery_duration) GROUP BY partner_id ORDER BY avg_time`.
- Revenue generated per partner — `SUM(order_amount) GROUP BY partner_id`.
- Vehicle-type performance and best partner per vehicle type — `GROUP BY vehicle_type; RANK() OVER (PARTITION BY vehicle_type ORDER BY avg_delivery_time)`.
- Overall performance ranking — composite ranking combining speed, volume, and rating using `DENSE_RANK()`.

Expected Output Description

A performance leaderboard ranking all 200 delivery partners by speed, delivery volume, revenue contribution, and customer rating, with a breakdown by vehicle type.

Business Insight

Electric bike and scooter-based partners show marginally faster average delivery times than bicycle-based partners in dense urban zones, while a small cohort of consistently top-ranked partners handle a disproportionate share of high-value deliveries, suggesting an opportunity to formalise an incentive tier for top performers.

Business Value

- Enable objective performance evaluation
- Support incentive and rewards planning
- Improve workforce optimisation

16. Advanced SQL Concepts Used

The project demonstrates SQL proficiency across beginner, intermediate, and advanced levels. The table below summarises the key concepts applied and their purpose within the project.

Level	Concepts	Application in Project
Basic	SELECT, WHERE, ORDER BY, LIMIT, DISTINCT	Filtering and retrieving core operational records
Intermediate	INNER/LEFT JOIN, GROUP BY, HAVING, Aggregate Functions, CASE	Combining tables and summarising business metrics
Advanced	Subqueries, CTEs, Window Functions (ROW_NUMBER, RANK, DENSE_RANK, PARTITION BY)	Multi-step revenue analysis, ranking, and segmentation
Date Functions	TIMESTAMPDIFF(), YEAR(), MONTH(), DAYNAME(), HOUR()	Delivery duration calculation and time-based trend analysis

Table 16.1: SQL Concept Summary

Query Distribution

Module	Number of Queries
Delivery Delay Analysis	10
Order Cancellation Analysis	10
Revenue Optimisation	10
Customer Retention Analysis	10
Delivery Partner Performance	10
Total	50

17. Business Insights Summary

Delivery

- Overall average delivery performance provides a baseline SLA benchmark.
- A defined set of restaurants and cities consistently underperform on delivery speed.
- Partner-level performance varies meaningfully, warranting targeted coaching.

Revenue

- Revenue is concentrated among top restaurants and cities.
- Digital payment methods are gaining share over cash-on-delivery.
- Monthly revenue trends reveal seasonal demand patterns.

Customers

- A premium customer segment drives a disproportionate share of revenue.
- A measurable share of customers have crossed the 90-day inactivity threshold.
- Segmentation enables tiered, personalised retention campaigns.

Delivery Partners

- Top performers can be identified and rewarded through an objective ranking system.
- Vehicle type shows a measurable, if modest, relationship with delivery speed.
- Underperforming partners can be identified early for support or retraining.

18. Power BI Dashboard Design

The SQL analysis outputs were connected to Power BI to build an interactive, multi-page dashboard aimed at non-technical executive stakeholders. The dashboard is organised into five pages, each aligned with one of the project's core business problem areas.

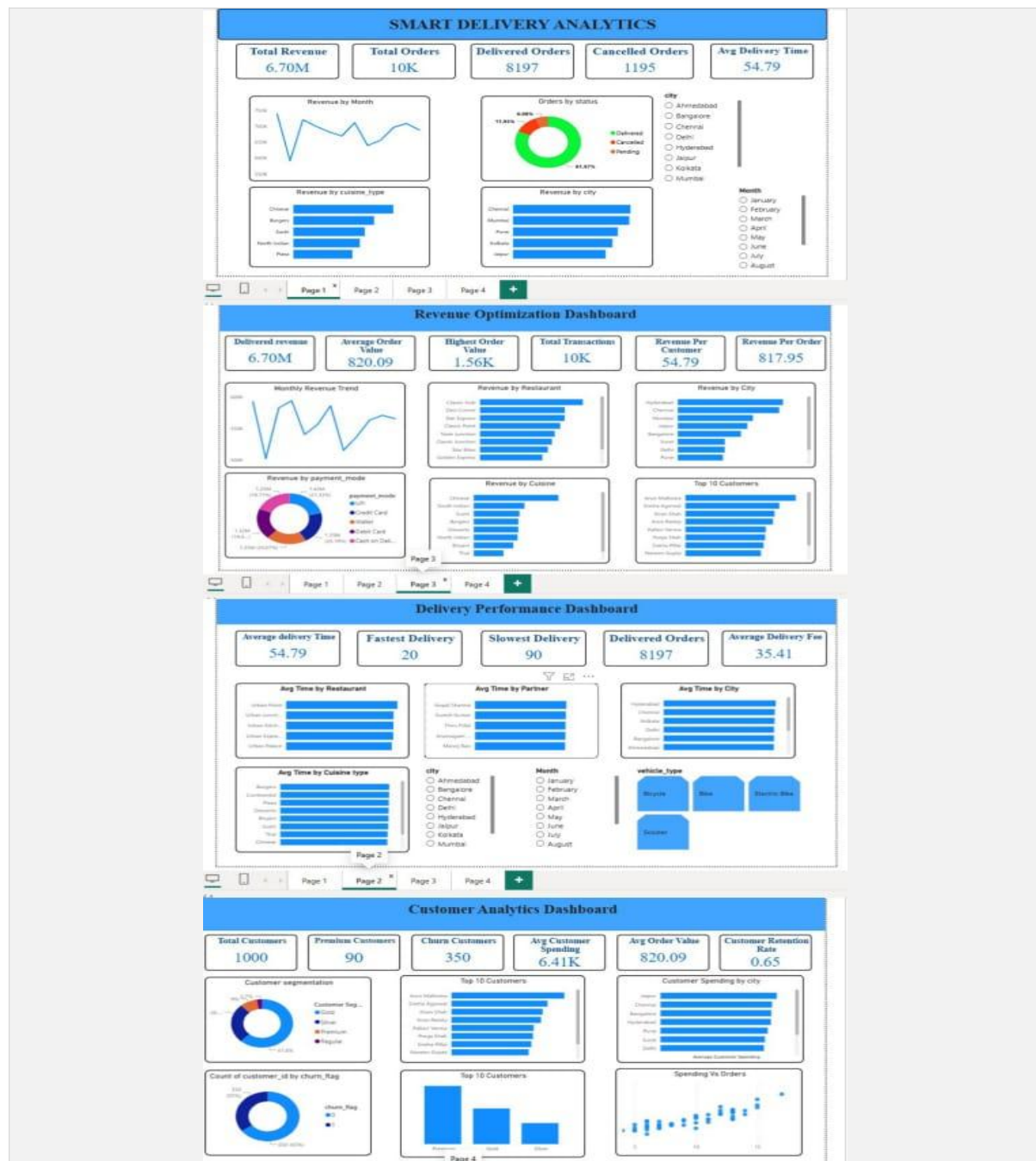


Figure 18.1: Power BI Dashboard Layout Placeholder

18.1 Executive KPI Overview

- Total Revenue, Total Orders, Cancelled Orders, Average Delivery Time, Customer Count

18.2 Revenue Dashboard

- Monthly Revenue Trend Chart
- Revenue by City and Cuisine
- Top Spending Customers

18.3 Delivery Dashboard

- City-wise Delivery Time Heat-map
- Restaurant Delivery Performance
- Top and Bottom Delivery Partners

18.4 Cancellation Dashboard

- Cancellation Rate KPI
- Cancellation by City and by Restaurant
- Cancellation Trend by Day and Hour

18.5 Customer Dashboard

- Customer Segmentation (Platinum/Gold/Silver/Bronze)
- Top Customers
- Churn Analysis View

19. Challenges Faced

The project encountered several real-world challenges typical of enterprise data analytics work, all of which were successfully resolved during implementation.

Challenge	Resolution
CSV import issues (encoding, delimiters)	Standardised file encoding and re-validated delimiters before LOAD DATA INFILE
Foreign key dependency errors during import	Enforced strict parent-before-child import order
Datetime conversion errors	Converted VARCHAR timestamps using ALTER TABLE MODIFY COLUMN after cleaning blanks
Blank / missing datetime values	Applied COALESCE() and conditional handling before conversion
Business rule validation failures	Added CHECK constraints and pre-load validation queries
Referential integrity verification	Used LEFT JOIN-based orphan-record detection queries
SQL query optimisation	Added indexes on frequently joined and filtered columns
Relationship / schema design decisions	Iteratively refined the ER diagram based on query requirements

20. Conclusion

This project successfully demonstrates the design and implementation of an end-to-end SQL-based analytics system for a simulated food delivery operation. Through the process of relational database design, systematic data cleaning, business rule validation, and referential integrity verification, a reliable analytical foundation was established for the dataset.

Fifty SQL queries spanning basic, intermediate, and advanced techniques — including joins, subqueries, CTEs, and window functions — were used to investigate five critical business problem areas: delivery delays, order cancellations, revenue optimisation, customer retention, and delivery partner performance. These findings were consolidated into an interactive Power BI dashboard that translates technical analysis into actionable insight for business stakeholders.

Overall, the project demonstrates the practical skill set expected of a professional Data Analyst: relational database design, SQL proficiency across all difficulty levels, rigorous data quality assurance, business-focused analytical thinking, and effective data storytelling through dashboards. The clean, validated dataset produced also establishes a strong foundation for future predictive analytics and machine learning initiatives.

21. Future Enhancements

- Predict customer churn using machine learning models such as Logistic Regression, Random Forest, or XGBoost.
- Predict delivery time based on historical patterns using regression and gradient boosting techniques.
- Build a restaurant recommendation engine based on customer ordering behaviour.
- Develop a real-time dashboard connected to live streaming operational data.
- Deploy the solution on a cloud platform such as AWS, Azure, or GCP.
- Automate data ingestion using ETL pipelines for continuous data refresh.

22. References

1. MySQL 8.0 Reference Manual — Oracle Corporation. Official documentation for MySQL database engine, SQL syntax, and window functions.
2. Microsoft Power BI Documentation — Microsoft Corporation. Official guidance on dashboard design, DAX, and data modelling.
3. Codd, E. F. "A Relational Model of Data for Large Shared Data Banks." Communications of the ACM, 1970. Foundational reference for relational database normalisation principles.
4. Industry case studies and publicly available analytics discussions on food delivery platform operations (Swiggy, Zomato, DoorDash, Uber Eats), used as contextual background for problem framing.
5. GitHub Documentation — Git and GitHub version control practices used for project documentation and script management.

23. Appendix

A. Sample CREATE TABLE Statement (Orders)

```
CREATE TABLE orders ( order_id INT PRIMARY KEY, customer_id INT, restaurant_id INT,
partner_id INT, order_time DATETIME, delivered_time DATETIME, order_amount DECIMAL(10,2)
CHECK (order_amount >= 0), delivery_fee DECIMAL(6,2) CHECK (delivery_fee >= 0), status
VARCHAR(20), FOREIGN KEY (customer_id) REFERENCES customers(customer_id), FOREIGN KEY
(restaurant_id) REFERENCES restaurants(restaurant_id), FOREIGN KEY (partner_id) REFERENCES
delivery_partners(partner_id) );
```

B. Sample Analytical Query (Delivery Partner Ranking)

```
WITH partner_stats AS ( SELECT partner_id, COUNT(order_id) AS total_deliveries,
AVG(TIMESTAMPDIFF(MINUTE, order_time, delivered_time)) AS avg_time FROM orders WHERE
status = 'Delivered' GROUP BY partner_id ) SELECT partner_id, total_deliveries, avg_time,
RANK() OVER (ORDER BY avg_time ASC) AS speed_rank FROM partner_stats;
```

C. List of Deliverables

- D1 — Database Design (ER Diagram)
- D2 — Table Creation Scripts (DDL)
- D3 — Data Import & Validation Log
- D4 — Data Cleaning Report
- D5 — SQL Analysis Queries (Minimum 40)
- D6 — Business Insights Report

Model Deployment: [link](#)